Conceptual Design Proposal

SDR-Based Monitoring System for Electric Field Exposure in High-Density IoT Indoor Environments

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Abstract

This document presents the conceptual design of an IoT platform based on Software-Defined Radio (SDR) for monitoring human exposure to electric fields in indoor environments with a high density of IoT devices. The proposed architecture consists of five main components: SDR hardware, a real-time acquisition engine in C, digital signal processing and business logic in Python, a TimescaleDB time-series database, and a reactive web interface. The system employs machine learning techniques for adaptive thresholding, cooperative spectrum sensing, and geostatistical interpolation (Kriging) for generating electromagnetic exposure heatmaps, all in compliance with ICNIRP and ITU-T K.52 standards. The hardware bill of materials (BOM), the RF signal chain, and the project timeline are included.

Keywords: SDR, IoT, electromagnetic exposure, Kriging, machine learning, USRP B210, Jetson Orin, heatmaps.

Contents

1 Introduction

1.1 Problem Statement

The proliferation of IoT devices in enclosed work environments (offices, factories, data centers) raises growing concern about cumulative exposure to electric fields. Traditional spectrum analyzers are expensive, bulky, and impractical for dense or mobile indoor studies. Software-Defined Radio (SDR) emerges as a lower-cost, more flexible alternative, albeit with limitations in dynamic range, ADC/DAC behaviour, and usable frequency band.

The system must operate in the proposed range of 700 MHz to 3500 MHz, covering the main IoT and telecommunications bands (LTE, WiFi 2.4 GHz, ISM). Indoor environments require numerous fast measurements to identify high-radiation zones and compare them against exposure standards.

1.2 Rationale

Static thresholding proves insufficient in noisy, dynamic IoT environments. The proposal incorporates adaptive thresholding based on machine learning and cooperative sensing. Electric field exposure maps allow correlating device density, frequency usage, and exposure levels, constituting both a scientific and a regulatory tool.

1.3 Research Question

How does IoT device density influence electric field exposure levels in enclosed environments, and what impact might such exposure have on workers' health, using an SDR-based monitoring system and advanced machine learning techniques for detection and analysis?

1.4 Objectives

General Objective Develop a system for assessing human exposure levels to electric fields, based on SDR and machine learning techniques, capable of measuring radiation density in indoor environments with high IoT device occupancy.

Specific Objectives

SO1: Design and build an SDR measurement module for spectral monitoring of high-density IoT networks in indoor spaces, using spectrum sensing techniques.

SO2: Develop thresholding algorithms and cooperative processes based on machine learning methods to handle measurement variability in indoor environments with high interference.

SO3: Generate spatial electric field maps using advanced measurement and data analysis techniques to assess compliance with human exposure regulations.

2 System Architecture

2.1 Overview

The system comprises five main modules working together to capture, process, store, and visualise electric field exposure data. The architecture is designed under the principles of real-time processing, low latency, and component decoupling, allowing each module to evolve independently.

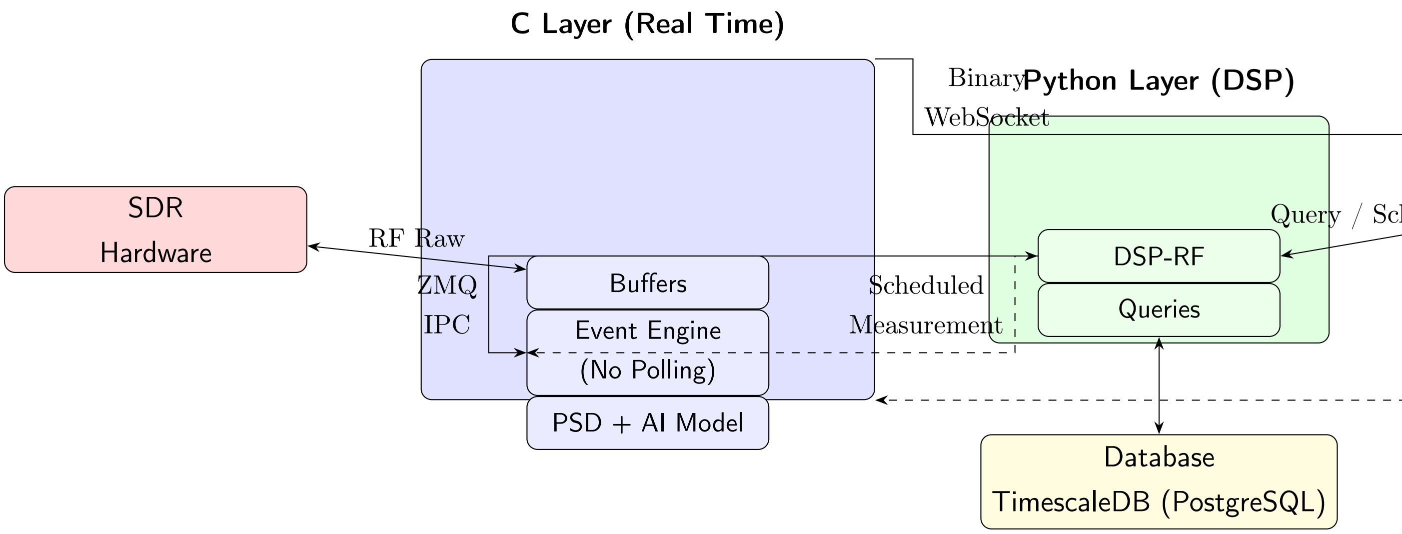

Figure 1: Conceptual diagram of the 5-component architecture. Solid lines represent primary data flows; dashed lines represent control commands.

2.2 Component 1: SDR (Software-Defined Radio)

Role: External RF capture device.

The SDR is the entry point for electromagnetic signals into the system. It captures raw RF samples, covering the range of 698 MHz to 6000 MHz with an instantaneous bandwidth of up to 56 MHz. The hardware selected for the prototype is the Ettus Research USRP B210, which offers a 2×2 MIMO architecture with 12-bit ADC/DAC resolution and a USB 3.0 interface.

The connection to the system is bidirectional: raw samples flow into the C-layer buffers, while configuration commands (frequency, gain, sampling rate) can be sent back to the SDR.

2.3 Component 2: C Layer — Real-Time Engine

Role: High-performance signal acquisition and preprocessing.

The C layer is the real-time processing core of the system. It is implemented in C11 and runs directly on the host hardware (not in a container), as it requires direct access to the SDR's USB peripherals. It consists of three submodules:

Buffers Act as a bridge between the SDR hardware and internal processing. They receive raw samples from the SDR in bidirectional mode and deliver them to the event engine, ensuring continuous flow without data loss.

Event Engine (No Polling) Event-driven architecture that explicitly avoids polling to minimise CPU usage and latency in real-time signal processing. It reacts to data arrival from the buffers and:

- Sends data and events to the Python layer via **ZMQ over IPC** (inter-process communication) for high-level processing.
- Receives control commands from Python (*scheduled measurement*).
- Streams binary spectral data directly to the web frontend via **WebSocket** (Libwebsockets), bypassing Python for latency-sensitive flows.
- Receives control commands from the frontend (*trigger*, *parameter changes*).

PSD (Power Spectral Density) + AI Model Power Spectral Density analysis module embedded directly in the C layer. It incorporates an artificial intelligence model for spectral analysis and anomaly detection, operating close to the data source for low-latency inference. The model is trained for adaptive thresholding and false-alarm reduction in high-interference environments.

2.4 Component 3: Python Layer — DSP and Business Logic

Role: High-level digital signal processing, data orchestration, and business logic.

The Python layer leverages the scientific ecosystem (NumPy, SciPy) for RF processing that does not require the minimal latency of the C layer. It consists of two submodules:

DSP-RF Digital Signal Processing module focused on RF. It receives events and data from the C layer via ZMQ IPC, applies advanced spectral analysis algorithms, and sends scheduled measurement commands back to the C layer. It also communicates bidirectionally with the frontend for *query* and *scheduling* operations for measurement campaigns.

Queries Data access layer that manages all persistence. It interacts bidirectionally with the database to store measurements, algorithm results, configurations, campaigns, and reports. It implements the business logic for scientific traceability: each measurement is linked to its acquisition configuration, AI model, campaign, and spatial location.

2.5 Component 4: Database — TimescaleDB

Role: Persistent time-series storage with analytical query capabilities.

TimescaleDB (an extension on top of PostgreSQL 16) provides:

- Efficient time-series storage (timestamped spectral measurements).
- Native compression and automatic partitioning (hypertables) for high-frequency data.
- Full relational model for metadata: sensors, antennas, configurations, campaigns, algorithms, thresholds, alerts, users, and compliance reports.
- End-to-end traceability from raw capture to exportable report.

2.6 Component 5: Frontend — User Interface

Role: Reactive visualisation for monitoring, control, and querying.

Single-page application (SPA) built with Vite + React + TypeScript, designed with Material UI (dark theme). The main visualisation components include:

- **Live Spectrum:** Waterfall plot and real-time FFT, fed directly by the binary WebSocket stream from the C layer (Plotly.js).
- **IoT Heatmap:** Electric field exposure map overlaid on floor plans, generated via Kriging interpolation from spatial measurements. Supports grid overlay and measurement point placement.
- **Sensor Status:** Health monitoring, telemetry, and connectivity of deployed SDR sensors.
- **Campaign Management:** Scheduling of measurement campaigns with sensor assignment, frequency ranges, resolution, and intervals.
- **Exportable Reports:** Automatic generation of regulatory compliance reports (ICNIRP, ITU-T K.52) with result caching.
- **Configuration Panel:** Service status, system parameters, and access control.

3 Communication Channels

Table 1 summarises the communication channels between components, the protocol used, and the data flow direction.

Table 1: System communication channels

Source	Destination	Protocol	Direction
SDR	Buffers (C)	Raw RF samples	Bidirectional
Buffers (C)	Event Engine (C)	Internal flow	Internal
Event Engine (C)	DSP-RF (Python)	ZMQ over IPC	Bidirectional
DSP-RF (Python)	Event Engine (C)	Scheduled measurement	Python → C
C Layer	Frontend (React)	Binary WebSocket	C → Frontend
Frontend (React)	C Layer	WebSocket (control)	Frontend → C
DSP-RF (Python)	Frontend (React)	HTTP (REST)	Bidirectional
Queries (Python)	Database	SQL	Bidirectional

3.1 Communication Design Decisions

ZMQ over IPC ZeroMQ with inter-process communication (IPC) provides low-latency messaging between the C and Python processes co-located on the same host, eliminating network stack overhead.

Direct binary WebSocket The spectral data stream is transmitted directly from the C layer to the browser without going through Python. This minimises latency for real-time visualisations (waterfall, FFT) that require a high refresh rate.

HTTP REST for queries Query and scheduling operations between the frontend and the Python backend use HTTP/REST, suitable for transactional operations that do not require continuous streaming.

4 Hardware Bill of Materials (BOM)

The monitoring system is portable and self-contained, designed to capture, process, and visualise RF signals in the range of 698 MHz to 6000 MHz. Each hardware component is detailed below with its role, specifications, supplier, and cost.

4.1 Cost Summary

Table 2: Hardware cost summary

Component	Unit Price (USD)	Imported Price (USD)
USRP B210	$2,200.00	$2,860.00
Antenna 698–2700 MHz	$200.00	$260.00
Preselection Filters (×2)	$200.00	$260.00
Jetson Orin Nano Super	$300.00	$390.00
7" Touchscreen	$60.00	$78.00
Custom Enclosure	$100.00	$130.00
TOTAL	**$3,060.00**	**$3,978.00**

4.2 Per-Component Specifications

4.2.1 SDR — Ettus Research USRP B210

Role: Main RF front-end. Broadband spectrum capture and transmission via USB 3.0 to the computing module.

Table 3: USRP B210 specifications

Parameter	Value
Model / Part	USRP B210 (Kit, UB210-KIT)
Supplier	Ettus Research (National Instruments)
Frequency range	70 MHz to 6000 MHz continuous
RF bandwidth	Up to 56 MHz instantaneous
Architecture	2×2 MIMO (2 TX, 2 RX)
ADC/DAC	12 bits
Host interface	USB 3.0
FPGA	Xilinx Spartan-6 XC6SLX150
Noise figure	≈8 dB
Impedance	50 Ω
RF connectors	4 SMA (RX/TX per channel)
Supported software	GNU Radio, UHD, MATLAB, LabVIEW

4.2.2 Computing Module — NVIDIA Jetson Orin Nano Super

Role: Central processing unit for AI inference, signal processing, system control, and display output.

Table 4: Jetson Orin Nano Super specifications

Parameter	**Value**
Model	Jetson Orin Nano Super Developer Kit (945-13766-0005-000)
AI performance	Up to 67 TOPS
GPU	NVIDIA Ampere, 1024 CUDA cores
CPU	6-core ARM Cortex-A78AE
RAM	8 GB LPDDR5, 102 GB/s bandwidth
Storage	2× M.2 Key M (NVMe SSD)
USB	4× USB 3.2 Gen 2 Type-A, 1× USB-C
Video	DisplayPort 1.2 (HDMI via adapter)
Connectivity	Wi-Fi (M.2 Key E), Gigabit Ethernet
GPIO	40-pin header
Operating system	JetPack (Ubuntu Linux)
AI support	TensorRT, PyTorch, ROS 2, GNU Radio

4.2.3 Omnidirectional Antenna — L-com LCANOM1075

Role: Main RF signal capture element in the 698 MHz to 2700 MHz range.

Table 5: Antenna specifications

Parameter	**Value**
Model	LCANOM1075
Manufacturer	L-com
Frequency range	698 MHz to 2700 MHz
Gain	6.5 dBi
Radiation pattern	Omnidirectional
Polarisation	Vertical
Connector	SMA Male (spring, 5.5 mm)
Radome	Fibreglass, black
Impedance	50 Ω
VSWR	$\leq$ 2.0 (typical)

4.2.4 Preselection Filters — Mini-Circuits

Role: Bandpass filtering between the antenna and the USRP B210. They suppress out-of-band interference before digitisation, improving sensitivity and reducing intermodulation. Two filters are used, one per RX channel of the USRP B210.

Table 6: Filter 1 — VBFZ-925-S+

Parameter	**Value**
Model	VBFZ-925-S+
Manufacturer	Mini-Circuits
Type	Connectorised bandpass filter
Passband	800 MHz to 1050 MHz
Connector	SMA
Impedance	50 Ω
Spectral coverage	GSM 850, GSM 900, LTE B5/B8/B20

Table 7: Filter 2 — BPF-A1340+

Parameter	**Value**
Model	BPF-A1340+
Manufacturer	Mini-Circuits
Type	LC bandpass filter
Passband	1000 MHz to 1800 MHz
Connector	SMA
Impedance	50 Ω
Spectral coverage	GPS L1 (1575 MHz), DCS 1800, LTE B1/B3/B4/B10

The combination of both filters provides preselection in the 800 MHz to 1800 MHz range, enabling simultaneous monitoring of two independent bands.

4.2.5 Touchscreen — Waveshare 7” HDMI LCD (C)

Role: Touch-based user interface for spectrum display, heatmaps, dashboards, and system status.

Table 8: Display specifications

Parameter	**Value**
Model	Waveshare 7" HDMI LCD (C), Rev 4.1
Size	7 inches
Panel type	IPS
Resolution	1024×600
Touch	Capacitive, 5-point
Video interface	HDMI
Touch interface	USB Micro-B (HID)
Supported systems	Ubuntu, Raspbian, Windows, JetPack

4.2.6 Custom Enclosure

Role: Mechanical container for all components. Provides physical protection, cable management, and mounting points for the antenna and display.

Design requirements:

- Accommodate USRP B210 PCB, Jetson Orin Nano, and 7 inch display.
- External antenna mounting port (SMA bulkhead).
- Access to USB 3.0 port for USRP B210 connection.
- Ventilation for thermal dissipation (Jetson $\approx 15\,\mathrm{W}$).
- Display window ($\approx 165\,\mathrm{mm} \times 100\,\mathrm{mm}$ visible area).
- Portable form factor, rugged instrument-case style.

4.3 RF Signal Chain

Figure 2 shows the RF signal flow from the antenna to the display.

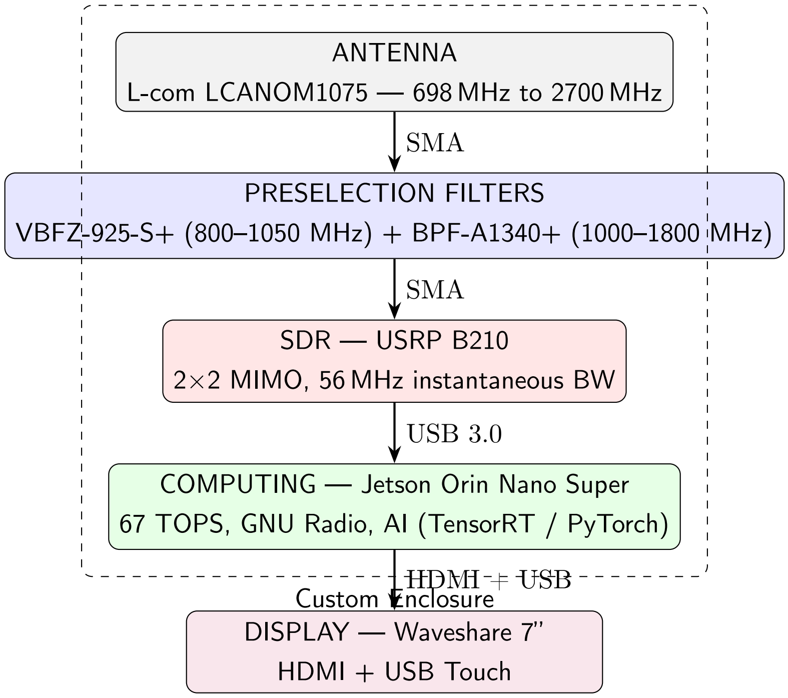

Figure 2: RF signal chain: from antenna to display.

5 AI and Interpolation Strategy

5.1 Adaptive Thresholding and Cooperative Sensing

The traditional static thresholding system is unsuitable for IoT environments with high device density, where noise and interference levels vary significantly. The proposal incorporates:

- **Autoencoders** for feature extraction and dimensionality reduction in raw spectral data. They enable identifying interference patterns characteristic of IoT environments and distinguishing them from anomalies requiring attention.
- **Two-level thresholding** inspired by cooperative communication systems (such as 5G): a fast first threshold for coarse detection and a refined second threshold, cooperatively validated across multiple sweeps or frequencies, for false-alarm reduction.
- **AI model embedded in the C layer** (PSD module) operating with low-latency inference, close to the data source, without requiring transfer to the Python backend for real-time decisions.

5.2 Kriging Interpolation

For generating spatial electric field exposure maps, Kriging interpolation is employed, a geostatistical method that:

- Estimates values at unsampled locations from a discrete set of measurement points with geographic coordinates (latitude, longitude).
- Provides not only the point estimate but also the estimation error variance, enabling assessment of the generated map's reliability.
- Reduces the number of required measurement points through intelligent selective sampling, preserving map quality and optimising the characterisation time of an indoor environment.

6 Conceptual Data Model

The data model is designed to ensure complete scientific traceability: every measurement can be traced back to its acquisition configuration, sensor, campaign, AI model, and location.

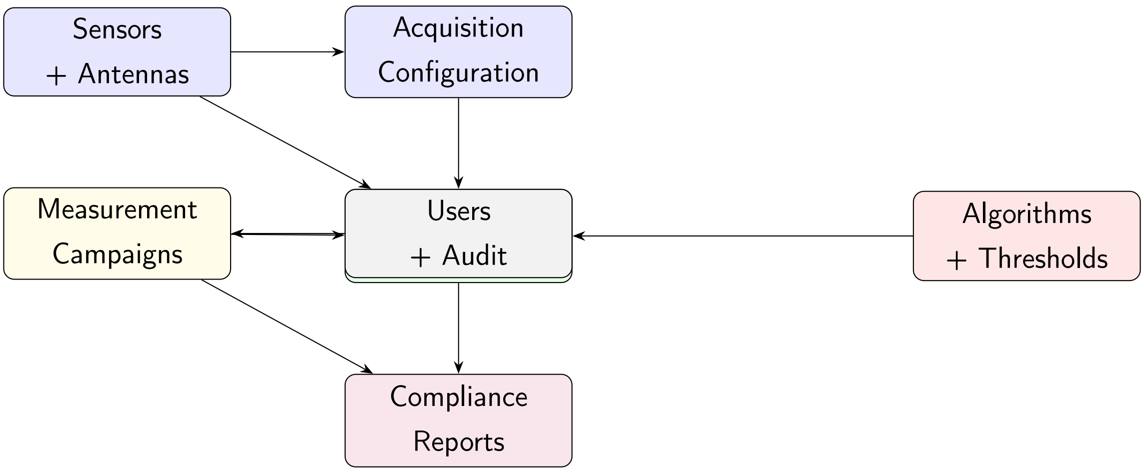

Figure 3: Conceptual data model: traceability from sensor to compliance report.

The main entities in the model include: sensors, antennas, acquisition configurations, measurement data (with spatial coordinates), campaigns, algorithms (with versions), configurable exposure thresholds, compliance reports, and audit logs.

7 Timeline

7.1 Execution Schedule (12 Weeks)

The practical execution plan is organised into three months with defined milestones:

Table 9: 12-week schedule

Month	Weeks	Key Activities
Month 1	1	Article writing; database audit; mathematical documentation of acquisition algorithms; initial SDR hardware setup.
	2	Historical data cleaning scripts; definition of thresholding parameters for IoT detection; sensor-to-database transfer rate validation.
	3	API/dashboard connection; Autoencoder environment for feature extraction; antenna adjustments and device positioning.
	4	First web visualisation tests with isolated data points; filtering tests with real data; SDR operational stability (TRL 7).
Month 2	5	Office heatmap UI design; Kriging interpolation implementation; systematic data collection.
	6	Autoencoder training with clean data; algorithm logic integration into visual platform; sensor calibration support.
	7	Dashboard optimisation for real-time rendering; mathematical model adjustment per ICNIRP standards; AI refinement to reduce false alarms.
	8	Full integrated test: capture $\rightarrow$ processing/AI $\rightarrow$ visualisation.
Month 3	9	Map accuracy validation against manual control measurements; automatic exportable reports; co-operative logic documentation.
	10	Prototype final assembly or disassembly; Autoencoder performance validation; bug fixes and access security.
	11	Final system tests; delivery of user manual and technical manual.
	12	Work plan closure, milestone review, and results presentation preparation.

7.2 Key Milestones

1. **Week 4:** Sensor data flowing to the platform in an organised manner.
2. **Week 8:** First functional version of the heatmap with Kriging interpolation.
3. **Week 12:** Operational, validated system with final deliverables.

8 Regulatory Compliance

The system is designed to assess electromagnetic exposure in accordance with:

- **ICNIRP** (International Commission on Non-Ionizing Radiation Protection): Guidelines for limiting exposure to electric and magnetic fields.
- **ITU-T K.52**: Guidance on complying with limits for human exposure to electromagnetic fields.
- **Colombian spectrum regulation** by the ANE (Agencia Nacional del Espectro — National Spectrum Agency).

Exposure thresholds are modelled as configurable domain data (not hard-coded values), enabling updates as standards evolve.

9 Expected Results

- Functional electric field exposure monitoring system validated in real environments at TRL 7.
- Integrated hardware and software prototype with a 5-component architecture.
- Adaptive thresholding and cooperative sensing algorithms with false-alarm reduction.
- Spatial exposure map generation via Kriging.
- At least one scientific publication indexed in a Minciencias-ranked journal.
- Software registration with the National Copyright Directorate (*Dirección Nacional de Derecho de Autor*).
- Public repository with source code, installers, manuals, and supporting documentation.
- Technical reports and thesis documents from affiliated students.

10 Participants

Table 10: Project team

Role	Name	Responsibility
Research group	GCPDS, UNAL Manizales	Coordination, schedule, integration
Principal investigator	Julio César García Álvarez	Scientific and technical leadership
Co-investigator	Marcelo Herrera González	Information analysis supervision
Undergraduate student	Alejandro Patiño Bedoya	Field data collection
Undergraduate student	Luis Felipe Giraldo Díaz	Technical reports and articles
Master's student	Julián Andrés Salazar Parias	AI algorithm design and optimisation

Supporting entities: Universidad Nacional de Colombia, Universidad de Caldas, Open Business Consulting S.A.S., Inference S.A.S., Dunderlab S.A.S.

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